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THE UNIVERSITY OF BRITISH COLUMBIA
UBC Sauder School of Business
Master of Business Analytics (MBAN)

BAIT 518 Data Visualization · Course Project Final Report

Follow the Complaints? Evaluating How New York City Targets Rat Inspections

An analysis of 311 rodent complaints and Department of Health rat inspections across 59 community districts, 2022–2025

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1. Introduction

You are a city rat inspector with one visit to make. East Harlem files more rodent complaints per resident than any other district (15.4 per 1,000 a year); Elmhurst–Corona in Queens files 1.8. Most people choose East Harlem. Yet inspectors find rats in 24% of East Harlem inspections and 55% of Elmhurst–Corona's. This report asks whether that mismatch is an exception or a pattern.

Most rat inspections in New York are triggered by 311 complaints, so where residents complain shapes where inspectors go, and therefore how much of the problem is found. Since 2017, DOHMH has also run Rat Mitigation Zones, where it inspects proactively instead of waiting for calls (Time Out New York, 2023). Our data cover four zones: Grand Concourse, East Village/Chinatown and Bedford-Stuyvesant/Bushwick, which predate our study period, and Harlem, designated in January 2023 (NYC DOHMH, n.d.-a).

Objective and audience. We ask whether complaint-driven targeting sends inspectors where rats are, and if not, where effort should shift. The audience is DOHMH Pest Control Services, the city's Director of Rodent Mitigation and City Council members who fund the zones; the decision supported is how to divide a fixed inspection budget across districts.

Answer in brief. Complaints do not point to the rats. Twelve districts are hidden hotspots, where rats are found often but complaints are rare. Queens has 27% of residents but receives 5.5% of inspections. Mitigation-zone districts reached a 15.5% found rate in 2025 against 21.0% elsewhere. We recommend that the city follow the rats, not the complaints, by shifting inspections toward districts with high found rates.

2. Data and Methodology

Sources. 311 Service Requests (Rodent) and Rodent Inspection (p937-wjvj) from NYC Open Data; DOHMH Rat Mitigation Zone indicators (validation); 2020 Census population from NYC City Planning's Community District Profiles; and DSNY monthly tonnage and DOHMH restaurant inspections, used to test whether trash or restaurant activity explains where rats are found. Full citations are in the References.

Preparation. A Python script (`clean_rat_data.py`) kept records within 2022–2025 with a valid community district (dropping 493 complaints and 16,300 inspections) and used only *initial* inspections, so re-visits are not double counted. Building-level results match inspections to properties by tax lot (BBL).

Metrics and district classification. The *rat found rate* is the share of initial inspections recording "Failed for Rat Activity"; it measures how often inspections identify signs of rat activity, not the number of rats. The *complaint rate* is rodent complaints per 1,000 residents per year (total complaints $\div$ 2020 district population $\times$ 1,000 $\div$ four years), so districts of different sizes can be compared. Districts are classified against the city medians of 4.15 complaints per 1,000 residents and a 23.3% found rate; a *hidden hotspot* has below-median complaints and an above-median found rate. For

districts outside the mitigation zones, the *priority score* averages their percentile ranks on both measures (0–100), with equal weight; higher scores mean higher priority.

Validation. Our rates match DOHMH's official zone rates within 0.2 to 2.2 percentage points on average across all 32 zone half-years; this is a consistency check, not proof that the methods are identical.

3. Results

3.1 Summer Brings More Eyes, Not More Rats

Annual complaints averaged roughly 38,000, with 42% filed between May and August. Yearly totals remained steady near 40,000 from 2022 through 2024. In 2025, complaints fell sharply to 31,300. Complaints range from 0.60 times the monthly average in December to 1.30 times in July, yet the rat found rate stays between 20% and 23% every month (Figure 1). Summer adds sightings, not infested buildings, so complaints measure attention rather than infestation. If complaints mislead about *when*, do they also mislead about *where*?

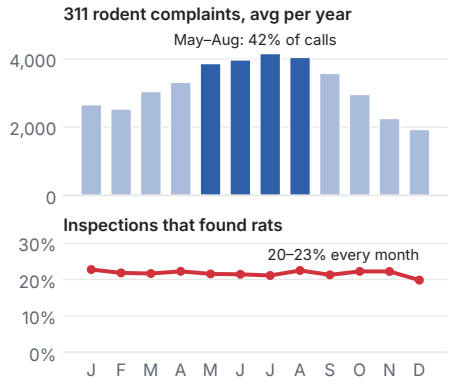


Figure 1: Average yearly complaints by month (top) and rat found rate by month (bottom), 2022–2025.

3.2 Twelve Districts Have Hidden Rat Problems

Twelve districts get fewer complaints than the median but have above-median found rates: five in Queens, four in Brooklyn, two in Manhattan and one in the Bronx (Figure 2). The rats are hiding where nobody is calling, so a complaint-driven system never sends anyone to look.Neither trash per resident nor restaurant rat violations explain them.

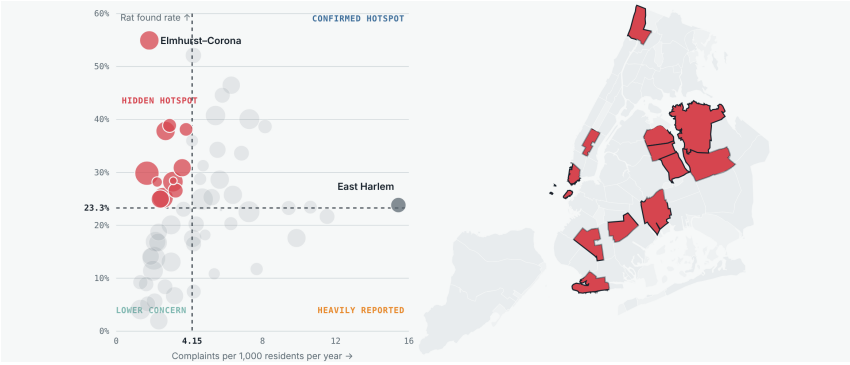


Figure 2: Complaint rate vs. rat found rate for 59 districts (left; circle size = population, dashed lines = city medians) and the twelve hidden hotspots on the map (right). From our presentation.

3.3 Queens Has 27% of New Yorkers but Gets 5.5% of Inspections

Queens holds 27.4% of residents and files 17.3% of complaints, but receives only 5.5% of inspections (Figure 3): one in four New Yorkers lives in Queens, but only one in eighteen inspections happens there. Yet it produces 7.4% of all rat findings.

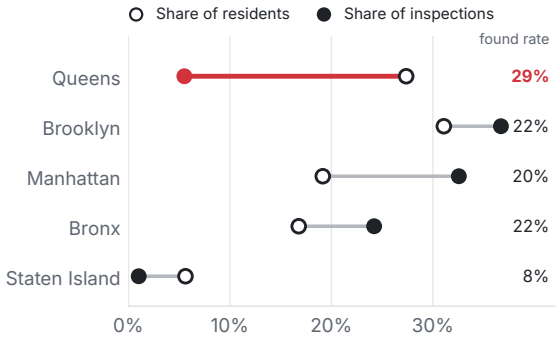


Figure 3: Share of residents (open) vs. share of inspections (filled) by borough, with each borough's found rate.

3.4 Half of All Rat Findings Come from One-Fifth of Buildings

Rats were found at 57,565 buildings. Over half had rats only once, but the repeat offenders carry the weight: the worst fifth of buildings, where rats were found again and again (every building with four or more findings, plus some with three), account for half of all findings, (Figure 4). Among those with a known type, 69% of the worst fifth are apartment buildings with three or more units (vs. 55% of all rat buildings), and one in three had no complaint matched to the property: complaints alone would miss them.

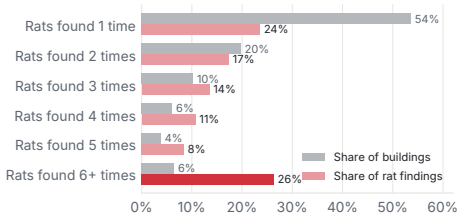


Figure 4: Share of buildings and of rat findings by number of times rats were found per building.

3.5 Mitigation Zones Pulled Ahead in 2025

Three of the four zones predate our study period; Harlem was added in January 2023. In 2022–2023, found rates inside and outside the zones were similar, at about 23%. By 2025, the zones had fallen to 15.5% against 21.0% elsewhere. The zones inspect every property, treat burrows and require sealed bins; in the West Harlem bin pilot, rat sightings fell 68%. This is consistent with targeted action working, though part of the gap reflects the zones inspecting more clean buildings.

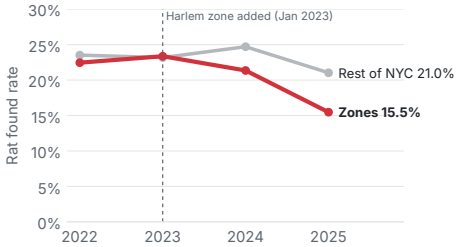


Figure 5: Rat found rate, mitigation-zone districts vs. rest of NYC. Harlem was added in January 2023.

4. Discussion and Limitations

Following complaints alone may mislead us about where rats are, leaving rat problems undetected in hidden hotspots.

Limitations:

- **Selection effects.** The data do not say whether an inspection was complaint-driven or proactive. Lightly inspected districts are likely inspected mostly after complaints, which can inflate their found rate, so the gap may be overstated though its direction should hold. *Check:* we compared inspections per complaint with found rate across districts, and re-ran the quadrants with ten different cut-offs. *Result:* complaint-driven districts do show higher rates (Greenpoint–Williamsburg: 1.2 inspections per complaint, 40.1%; Bedford-Stuyvesant: 13.6, 17.6%), but 9 of the 12 hidden hotspots stay hidden in every version, and 11 of 12 sit above the median within 95% confidence intervals.
- **Zone results are correlational.** Zones inspect clean properties too, which lowers the rate mechanically, and three zones have no pre-zone baseline in our data. *Check:* our zone rates move in the same direction as DOHMH's published zone figures, and city data on the same period show rat sightings fell 68% in the West Harlem bin pilot, against 16% in other zones and 6% citywide. We read the zone gap as strong support, not proof.
- **Scope.** 2022–2025 only; 2020 Census population held constant; the 2025 drop in complaints is not explained by our data. *Check:* totals reconcile by year and by district, and district populations sum to within 1,000 of the 2020 Census (8,803,220 vs. 8,804,190).

5. Recommendations

Based on our priority scoring, each district receives a single designated action (Figure 6). The framework maintains the 9 existing mitigation zones and designates 5 new zone candidates. For the remainder, it adds proactive inspections across 22 districts, primarily in Brooklyn and Queens. The ten highest-priority districts outside today's zones are listed in Table 1; the top five find rats in 39–46% of inspections, and Flatbush–Midwood, Sunset Park and Sunnyside–Woodside also have low inspection coverage.

Table 1: Top ten priority districts outside existing Rat Mitigation Zones

Rank	District	Action	Score	Found rate	Complaints/1k/yr	Inspections/1k/yr
1	Greenpoint–Williamsburg (BK 1)	New zone	92	40.1%	7.3	8.7
1	Park Slope–Red Hook (BK 6)	New zone	92	38.7%	8.1	7.2
3	Flatbush–Midwood (BK 14)	New zone	91	46.5%	6.3	3.8*
4	Sunset Park (BK 7)	New zone	89	44.6%	5.8	4.8*
5	Astoria (QN 1)	New zone	84	40.7%	5.4	6.0
6	Downtown Brooklyn–Fort Greene (BK 2)	Proactive	83	33.6%	6.9	8.8
7	Sunnyside–Woodside (QN 2)	Proactive	79	52.1%	4.2	4.3*
8	Kingsbridge Hts–Bedford Park (BX 7)	Proactive	78	34.3%	5.5	30.3
9	Crown Heights North (BK 8)	Proactive	74	23.4%	10.6	54.4
10	Washington Hts–Inwood (MN 12)	Proactive	73	28.6%	5.7	18.8

Note. Score = priority score (0–100). New zone = candidate Rat Mitigation Zone; Proactive = add proactive inspections. * Below the city median inspection rate.

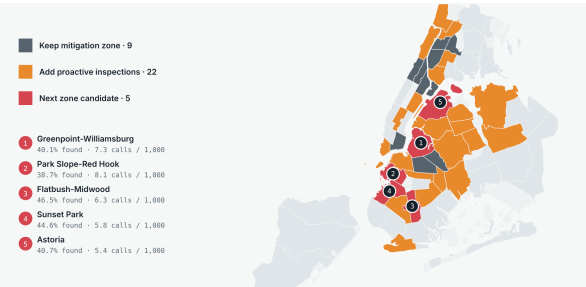


Figure 6: Recommended action for every district; the five next-zone candidates are numbered. From our presentation.

Five Recommendations for New York.

1. **Rank inspections by rats found, not complaints.** Complaints swing with the seasons and miss the 12 hidden hotspots.
2. **Send proactive inspectors to Queens,** where 27% of residents live but 5.5% of inspections happen.
3. **Launch five new mitigation zones,** starting with the repeat-offender buildings.
4. **Fund the expansion:** at Harlem's $3.5 million first-year cost, five zones need roughly $17.5 million.
5. **Build a photo-report app:** residents photograph burrows, droppings or live rats with the location, so inspectors can check the evidence and rank reports before they go (our proposal; untested).

6. Conclusion and Next Steps

Complaints tell you where people are looking; data tells you where the rats are. Shifting part of the inspection budget toward evidence-based targeting would put limited capacity where it is most needed.

The project opened more questions than a data-visualization course can answer. Due to course constraints on time and data, several deeper analyses were beyond our scope. These include separating complaint-driven from proactive inspections, isolating causal zone effects, a fuller analysis of building and waste data, and extending the timeline prior to 2022. We intend to keep pursuing the question and to turn it into a real business project. One direction, suggested by Professor Chunhua after our presentation, is to apply the same approach to rodent problems on the UBC campus with UBC Sauder.

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A. Presentation Materials, Submitted Files and Reproduction Guide

Our presentation combined several formats, all built from the same cleaned data:

- **Interactive opening** (“You are the inspector”): the audience picks between East Harlem and Elmhurst–Corona before the answer is revealed (Figure 7).
- **Animated chart scenes**: nine scenes that build step by step, from the overview numbers through seasonality, hidden hotspots, Queens, mitigation zones, repeat-offender buildings, recommended actions and the five moves (Figures 2 and 6 are taken from these scenes).
- **Interactive rat animation**: inside the hidden-hotspot scene, cartoon rats swarm the twelve hidden-hotspot districts while a puzzled inspector stands where the complaints point (Figure 8).
- **3D walkthrough model** (*Follow the Rats*): a sandbox-style city in which an inspector walks past 3D versions of each chart, one stop per finding (Figure 9).
- **NYC Rat Atlas**: an interactive map of all 59 districts for exploring the data after the talk.

Online version. The interactive opening, including the rat animation, can be viewed at `https://claude.ai/artifact/6ma5XqUDRxZt1684pLSDnH`; its “See the data” button leads to the animated chart scenes.

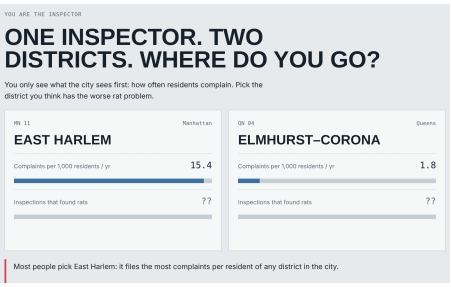


Figure 7: Interactive opening.

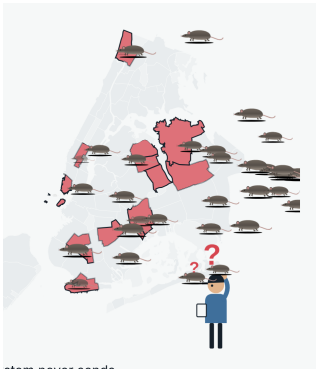


Figure 8: Interactive rat animation.



Figure 9: *Follow the Rats* 3D model.

The raw downloads are too large to submit (the inspection file alone is about 860 MB) and can be obtained from the sources in the References. Running the three scripts in order reproduces every table and figure.

Table 2: Submitted files

File	Description and use
`clean_rat_data.py`	Step 1. Cleans raw downloads; writes `tableau_data/` and `cleaning_log.txt`.
`analyze_rat_data.py`	Step 2. Exploratory analysis; writes `tableau_data/analysis/`.
`build_presentation_data.py`	Step 3. One table per final chart plus priority scores; writes `presentation_data/`.
`tableau_data/`, `presentation_data/`	Cleaned and chart-ready tables; join on `boro_cd`.
`group project.twbx`	Tableau packaged workbook (our own charts).
`Presentation Group 7/`	Interactive opening and animated chart scenes (HTML; arrow keys step through builds).
`Follow the Rats.html`	3D walkthrough model (open in a browser with internet access).
`nyc-rat-atlas/index.html`	Interactive district map; self-contained.
AI prompt history	Complete conversation exports (see Appendix B).

B. Use of Generative AI

In line with course guidance, the team used Claude (Anthropic) and ChatGPT (OpenAI) throughout the project. These tools helped us search for and evaluate data sources; write the Python scripts for cleaning, analysis and chart preparation; check figures, including the validation against DOHMH's official zone rates; and draft and format this report. We also used them to turn the charts we first built ourselves in Tableau into smooth animated chart builds for the presentation, and to create the interactive rat animation, the *Follow the Rats* 3D sandbox model and the NYC Rat Atlas.

Throughout, we kept our own critical judgement. We chose the topic, the question and the story, and we treated everything the AI searched for or produced as a draft. We cut, adjusted and rewrote much of it, checked whether each claim was accurate against the underlying data or its original source, and judged whether it served our purpose and audience. For example, while finalising this report we checked the history of the Rat Mitigation Zones against DOHMH's own records and corrected an earlier claim that all four zones were launched in 2023. All figures in this report are produced by the submitted scripts and can be reproduced.
The complete prompt history is submitted with this report:

- Claude conversations: `https://claude.ai/share/faff68b5-934b-4657-9019-09d5dfa26e79`
- ChatGPT conversation: `https://chatgpt.com/s/cx_6ae562defd2c8191aadbd87ad2efa80a`